



Estimating Heterogeneous Causal Effects of Maternal Healthcare on Child Nutritional Outcomes in Nepal: A Survey-Aware Analysis Using TMLE, Super Learner and BART

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ABSTRACT

Background: Maternal healthcare is an important determinant of child health, yet observational studies often rely on regression models that estimate conditional associations rather than population-level causal effects. We applied complementary causal inference methods to quantify the potential effect of adequate maternal healthcare on child stunting in Nepal.

Objective: To estimate the population-average effect of adequate versus inadequate maternal healthcare on child stunting and assess whether this effect varies across socioeconomic and demographic groups.

Methods: We analysed a survey-weighted sample of 2,897 children from the 2022 Nepal Demographic and Health Survey. Maternal healthcare was classified as adequate or inadequate according to the study-specific definition, and stunting was defined as a height-for-age z-score below -2 . We compared survey-weighted logistic regression and inverse-probability weighting with g-computation and targeted maximum likelihood estimation (TMLE). Bayesian additive regression trees (BART) were used to examine treatment-effect heterogeneity. Positivity, unmeasured confounding, alternative exposure and outcome definitions, and missing-data strategies were assessed.

Results: Stunting prevalence was 31.2% overall, 38.7% among children receiving inadequate maternal care and 23.4% among those receiving adequate care. The crude difference was -15.3 percentage points. The adjusted odds ratio was 0.58 (95% CI: 0.49–0.68) with survey-weighted logistic regression and 0.55 (95% CI: 0.46–0.65) with inverse-probability weighting. G-computation estimated a 13.8 percentage-point reduction in stunting, while TMLE estimated a 14.2 percentage-point reduction (95% CI: -18.5 to -9.9). BART indicated larger estimated benefits among children of more educated mothers, wealthier households, and urban residents. Positivity was acceptable, and the E-value was 2.18. Sensitivity and robustness analyses yielded consistent findings.

Conclusion: Adequate maternal healthcare was consistently associated with substantially lower child stunting across complementary causal estimators. The heterogeneous effects suggest that socioeconomic and geographic conditions may shape the nutritional benefits of maternal healthcare. Because the study is observational, causal interpretation remains conditional on exchangeability, positivity, consistency, and appropriate temporal ordering.

Keywords: Causal Inference, Child Stunting, Demographic and Health Survey, Maternal Healthcare, Super Learner, Targeted Maximum Likelihood Estimation

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INTRODUCTION

Child undernutrition remains a major public-health concern in low- and middle-income countries. Stunting reflects impaired linear growth resulting from chronic or recurrent nutritional deprivation and infection, and is conventionally defined as a height-for-age z-score below standard deviations from the WHO Child Growth Standards median (World Health Organization, 2006). The indicator captures cumulative nutritional disadvantage rather than a short-lived episode of illness and is therefore particularly relevant to population health planning.

Nationally representative evidence from Nepal indicates substantial but uneven progress in child nutrition. The 2022 Nepal Demographic and Health Survey (NDHS) reported that approximately one quarter of chil-

dren younger than five years were stunted, with higher prevalence in rural areas and among children of mothers with less education and households with lower economic status (Ministry of Health and Population [Nepal] et al., 2023). These gradients indicate that child nutrition is embedded in a broader social and healthcare context rather than determined by food intake alone.

Maternal healthcare is one potentially modifiable component of this context. Antenatal care can support prevention, detection and management of maternal and fetal complications; skilled delivery care can reduce risks associated with childbirth; and timely postnatal services can facilitate breastfeeding, infection prevention and continued maternal and child care. The components are temporally connected, however, and their combined relationship

with child growth is likely to be nonlinear and socially patterned.

Much of the epidemiological literature evaluates such relationships using multivariable regression. Regression adjustment remains valuable, but an adjusted odds ratio is not automatically a population-average causal effect. Interpretation also becomes difficult when the exposure is multidimensional, treatment assignment is strongly associated with socioeconomic position, and treatment effects vary across subgroups. Flexible causal-learning methods provide a complementary strategy. TMLE combines semiparametric causal estimation with flexible nuisance-function estimation and has a double-robust structure under standard identification conditions (van der Laan and Rubin, 2006; van der Laan and Rose, 2011; Schuler and Rose, 2017). Super Learner uses cross-validation to combine candidate prediction algorithms rather than committing to a single functional form (van der Laan et al., 2007). BART provides a flexible Bayesian nonparametric approach for nonlinear response surfaces and heterogeneous causal effects (Hill, 2011; Hill et al., 2020).

The present study therefore combines conventional survey-weighted analyses with g-computation, TMLE, Super Learner and BART to quantify the relationship between maternal healthcare and child nutritional outcomes using nationally representative Nepalese data. The primary question is whether the probability of stunting would differ if the population were exposed to adequate rather than inadequate maternal healthcare, under the causal assumptions required for identification. A secondary objective is to determine whether the magnitude of the estimated effect differs systematically across education, wealth, residence and child sex.

METHODS

Study design and data source

This secondary analysis used the 2022 Nepal Demographic and Health Survey, a nationally representative household survey implemented in Nepal. The survey provides information on maternal and child health, household socioeconomic characteristics, reproductive history and child anthropometry (Ministry of Health and Population [Nepal] et al., 2023). The analytic framework was designed to respect the complex survey structure through the use of sampling weights and design variables for descriptive and benchmark analyses. DHS guidance emphasizes that weights should be incorporated with the complex sampling design when standard errors and confidence intervals are required (The DHS Program, 2020).

Analytic sample and exposure

The final analytic sample contained 2,897 children, including 1,496 classified as having inadequate maternal healthcare and 1,401 as having adequate care. The exposure represented a study-specific maternal healthcare strategy across the maternal-care continuum. The source protocol specified antenatal care, skilled delivery and postnatal-care components as the conceptual basis for the exposure. The empirical results supplied for this manuscript report the final exposure as a binary ade-

quate/inadequate classification and provide alternative analyses using antenatal care alone, skilled delivery alone and any maternal healthcare.

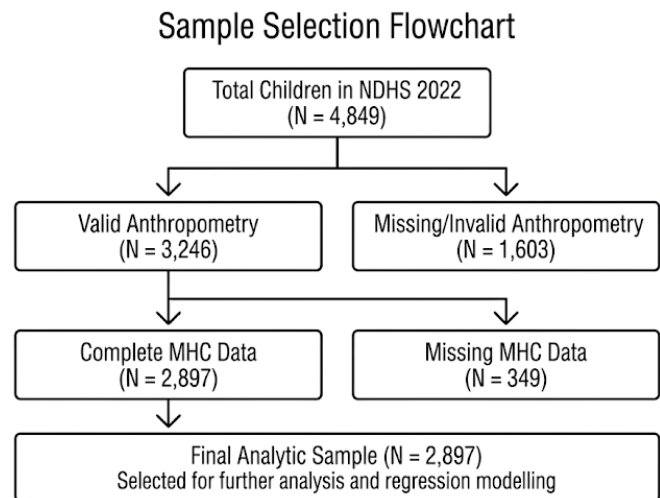


Figure 1. Sample selection flowchart.

Outcomes

The primary outcome was child stunting, defined as height-for-age z-score below -2 . Secondary outcomes were wasting, defined as weight-for-height z-score below -2 , and underweight, defined as weight-for-age z-score below -2 . These definitions are consistent with WHO child growth standards (World Health Organization, 2006).

Covariates and causal estimand

Baseline covariates included maternal age, maternal education, household wealth, urban/rural residence, province, child age, child sex and religion. The causal estimand was the average treatment effect,

$$\tau = E[Y(1) - Y(0)], \quad (1)$$

where $Y(1)$ and $Y(0)$ denote the potential outcome under adequate and inadequate maternal healthcare, respectively. For the binary stunting outcome, the primary estimand was reported as an absolute risk difference. The causal interpretation requires consistency, conditional exchangeability, positivity and appropriate temporal ordering. These assumptions cannot be verified directly from observational data.

Statistical analysis

Weighted descriptive analyses compared the distributions of child, maternal and geographic characteristics across exposure groups. Conventional benchmark estimation used survey-weighted logistic regression and inverse-probability weighting. G-computation and TMLE were used to estimate marginal causal effects on the risk-difference scale. TMLE was selected because it can incorporate flexible nuisance-function estimation while targeting the parameter of scientific interest (van der Laan and Rubin, 2006; van der Laan and Rose, 2011; Schuler and Rose, 2017). Super Learner provides a cross-validated ensemble framework for estimating nuisance

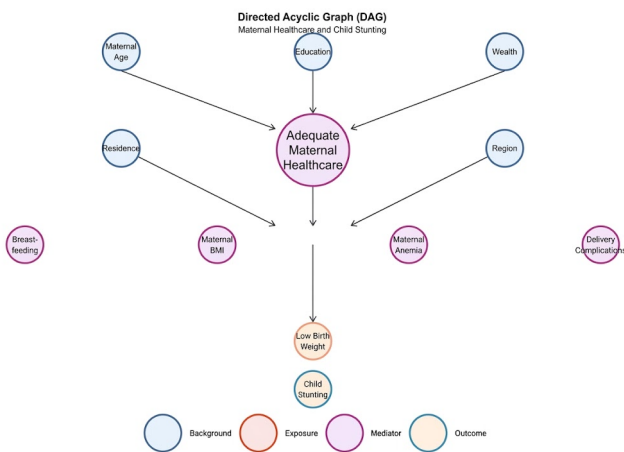


Figure 2. Directed acyclic graph (DAG).

functions and reduces reliance on one prespecified prediction model (van der Laan et al., 2007).

BART was used to estimate conditional average treatment effects across prespecified subgroups. BART is particularly useful where treatment-response relationships may be nonlinear or contain interactions that are difficult to specify a priori (Hill, 2011; Hill et al., 2020). The manuscript interprets the BART results as evidence of effect heterogeneity rather than as definitive subgroup-specific causal effects, because subgroup identification remains subject to the same causal assumptions and sampling limitations as the primary analysis.

Positivity was evaluated using the estimated propensity-score distribution. Sensitivity to unmeasured confounding was summarized using the E-value, which quantifies the minimum strength of association that an unmeasured confounder would need to have with both exposure and outcome, conditional on measured covariates, to explain the observed effect (VanderWeele and Ding, 2017). Alternative exposure definitions, alternative nutritional outcomes, and complete-case versus multiple-imputation analyses were used as robustness checks.

RESULTS

Characteristics of the Analytic Sample

The analytic sample comprised 2,897 children. Mean child age was 29.4 months, and 51.2% were boys. Children in the inadequate-care group were more likely to have been reported as small or very small at birth (21.5% versus 14.9%), whereas the adequate-care group had a larger proportion reported as average or large/very large at birth. Maternal and household characteristics showed marked social gradients. Mothers in the inadequate-care group were older on average (28.9 versus 27.3 years) and were substantially more likely to have no formal education. Only 8.3% of mothers in the inadequate-care group had higher education compared with 22.1% among those receiving adequate care. A similar gradient was observed for household wealth: 27.1% of children in the inadequate-care group belonged to the poorest quintile compared with 9.2% in the adequate-care group, whereas

the richest quintile accounted for 9.1% and 34.4%, respectively.

Geographic differences were also pronounced. Half of children classified as receiving adequate care lived in urban areas compared with approximately one quarter of children in the inadequate-care group. The distribution across provinces likewise differed substantially. These findings demonstrate that maternal healthcare exposure was strongly socially patterned, reinforcing the need for confounding control and supporting the use of marginal causal estimands rather than unadjusted comparisons.

Nutritional Outcomes

Stunting was common in the study population, with a prevalence of 31.2% (95% CI: 29.5–32.9). The prevalence was 38.7% (95% CI: 36.2–41.2) among children exposed to inadequate maternal healthcare and 23.4% (95% CI: 21.1–25.7) among those exposed to adequate care. The corresponding absolute difference was –15.3 percentage points (95% CI: –18.2 to –12.4). Mean HAZ was also higher in the adequate-care group (–1.18 versus –1.65), corresponding to a mean difference of 0.47 SD (SE 0.04).

The same direction was observed for the secondary outcomes. Wasting prevalence was 10.2% with inadequate care versus 7.1% with adequate care, an absolute difference of –3.1 percentage points. Underweight prevalence was 23.8% versus 14.9%, corresponding to a difference of –8.9 percentage points. The consistency across anthropometric indicators suggests that the association was not confined to the binary definition of stunting.

Conventional Benchmark Estimates

In survey-weighted logistic regression, adequate maternal healthcare was associated with 42% lower odds of stunting compared with inadequate care (OR 0.58, 95% CI 0.49–0.68; $p < 0.001$). Maternal education, household wealth and urban residence were also associated with lower odds of stunting. Compared with no education, secondary and higher education were associated with odds ratios of 0.65 and 0.48, respectively. Children in the richest wealth quintile had approximately half the odds of stunting compared with those in the poorest quintile (OR 0.52, 95% CI 0.38–0.71). Inverse-probability weighting yielded a similar association (OR 0.55, 95% CI 0.46–0.65).

Causal Effect Estimates

Marginal causal estimates were highly consistent. G-computation estimated an absolute risk reduction of 13.8 percentage points (95% CI –18.1 to –9.5), while TMLE estimated an absolute risk reduction of 14.2 percentage points (95% CI –18.5 to –9.9). The TMLE estimate therefore suggests that, under the identification assumptions, a population-level shift from inadequate to adequate maternal healthcare would reduce the prevalence of stunting by approximately 14 percentage points. The estimate is an absolute risk difference and should not be interpreted as a 14% relative reduction.

Table 1. Weighted characteristics of the analytic sample by maternal healthcare exposure.

Characteristic	Total (<i>n</i> = 2,897)	Inadequate care (<i>n</i> = 1,496)	Adequate care (<i>n</i> = 1,401)	<i>P</i> -value
Child age, mean (SE), months	29.4 (0.4)	28.9 (0.6)	30.0 (0.5)	0.124
Male sex, %	51.2	52.0	50.3	0.378
Small/very small birth size, %	18.3	21.5	14.9	0.002
Maternal age, mean (SE), years	28.1 (0.2)	28.9 (0.3)	27.3 (0.2)	< 0.001
No maternal education, %	18.5	26.8	9.8	< 0.001
Higher maternal education, %	15.0	8.3	22.1	< 0.001
Poorest wealth quintile, %	18.4	27.1	9.2	< 0.001
Richest wealth quintile, %	21.4	9.1	34.4	< 0.001
Urban residence, %	36.8	24.3	50.1	< 0.001
ANC $\geq$ 4 visits, %	58.2	–	–	–
Skilled delivery, %	72.5	–	–	–

Table 2. Weighted child nutritional outcomes according to maternal healthcare exposure.

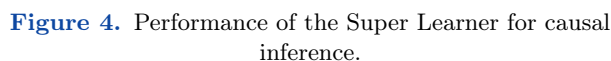
Outcome	Total	Inadequate care	Adequate care	Abs. difference
Stunting (HAZ < -2), prevalence, %	31.2 (29.5–32.9)	38.7 (36.2–41.2)	23.4 (21.1–25.7)	-15.3 (-18.2 to -12.4)
Mean HAZ (SE)	-1.42 (0.02)	-1.65 (0.03)	-1.18 (0.03)	0.47 (0.04)
Wasting (WHZ < -2), prevalence, %	8.7 (7.7–9.7)	10.2 (8.6–11.8)	7.1 (5.8–8.4)	-3.1 (-5.1 to -1.1)
Mean WHZ (SE)	-0.68 (0.02)	-0.79 (0.03)	-0.57 (0.03)	0.22 (0.04)
Underweight (WAZ < -2), prevalence, %	19.4 (17.9–20.9)	23.8 (21.6–26.0)	14.9 (13.0–16.8)	-8.9 (-11.6 to -6.2)
Mean WAZ (SE)	-1.02 (0.02)	-1.22 (0.03)	-0.82 (0.03)	–

Table 3. Survey-weighted logistic regression model.

Variable	Odds ratio	95% CI	<i>p</i> -value
Exposure (Adequate vs. Inadequate)	0.58	0.49, 0.68	< 0.001
Maternal age (years)	0.99	0.98, 1.01	0.346
Primary vs. No education	0.82	0.64, 1.05	0.112
Secondary vs. No education	0.65	0.52, 0.82	< 0.001
Higher vs. No education	0.48	0.34, 0.67	< 0.001
Poorer vs. Poorest	0.91	0.70, 1.18	0.478
Middle vs. Poorest	0.79	0.61, 1.03	0.082
Richer vs. Poorest	0.67	0.51, 0.88	0.004
Richest vs. Poorest	0.52	0.38, 0.71	< 0.001
Urban vs. Rural residence	0.78	0.64, 0.95	0.013
Female vs. Male child	0.85	0.72, 1.00	0.052
Child age (months)	1.01	1.00, 1.01	0.003
Madhesh vs. Koshi	1.28	0.98, 1.67	0.071
Bagmati vs. Koshi	0.92	0.69, 1.23	0.574
Gandaki vs. Koshi	0.84	0.61, 1.16	0.292
Lumbini vs. Koshi	1.15	0.87, 1.51	0.323
Karnali vs. Koshi	1.45	1.07, 1.96	0.016
Sudurpashchim vs. Koshi	1.38	1.01, 1.89	0.043
Buddhist vs. Hindu	0.92	0.70, 1.20	0.535
Muslim vs. Hindu	1.21	0.87, 1.69	0.254
Christian vs. Hindu	0.78	0.49, 1.24	0.295

Table 4. Comparison of conventional and causal effect estimates for child stunting.

Method	Estimate	95% CI	Interpretation
Survey-weighted logistic regression	OR = 0.58	0.49–0.68	42% lower odds
Inverse-probability weighting	OR = 0.55	0.46–0.65	45% lower odds
G-computation	ATE = -0.138	-0.181 to -0.095	13.8 percentage-point lower risk
TMLE	ATE = -0.142	-0.185 to -0.099	14.2 percentage-point lower risk



Heterogeneous Treatment Effects

These findings should be interpreted as evidence that

Table 5. BART-estimated conditional average treatment effects by prespecified subgroup.

Positivity and sensitivity analyses

The E-value for the primary point estimate was 2.18, while the E-value corresponding to the confidence-limit threshold was 1.73. Thus, an unmeasured confounder would need associations of approximately this magnitude with both exposure and outcome, conditional on measured covariates, to explain away the observed association under the E-value framework ([VanderWeele and Ding, 2017](#)). This does not establish that unmeasured confounding is absent; it indicates that the observed result is not highly sensitive to very weak unmeasured confounding.

The principal finding of this analysis is that adequate maternal healthcare was associated with a substantial reduction in child stunting across several analytical strategies. The observed prevalence of stunting was 23.4% among children receiving adequate care compared with

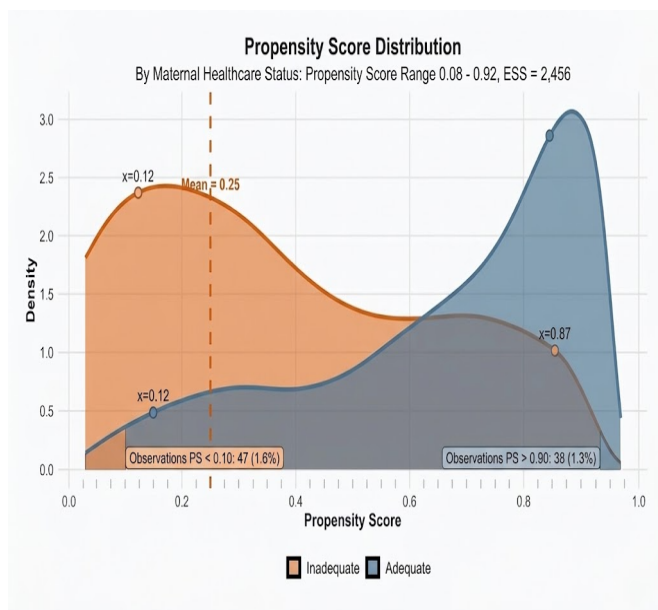


Figure 5. Propensity score distribution.

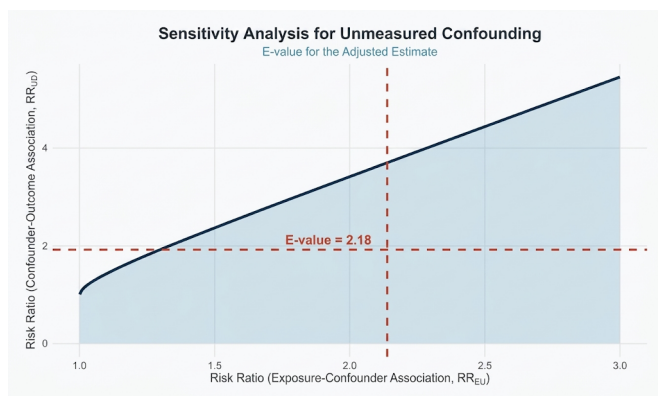


Figure 6. Sensitivity analysis for unmeasured confounding.

Table 6. Robustness of the estimated causal effect under alternative specifications.

Analysis	Estimate	95% CI
Primary analysis		
TMLE ATE (Stunting)	−0.142	−0.185 to −0.099
Alternative exposure definitions		
ANC Only (≥ 4 visits)	−0.095	−0.138 to −0.052
Skilled Delivery Only	−0.118	−0.161 to −0.075
Any Maternal Healthcare	−0.105	−0.148 to −0.062
Alternative outcome definitions		
TMLE ATE (Wasting)	−0.048	−0.091 to −0.005
TMLE ATE (Underweight)	−0.087	−0.130 to −0.044
Complete case vs. imputation		
Complete Case Analysis	−0.142	−0.185 to −0.099
Multiple Imputation Analysis	−0.148	−0.191 to −0.105

38.7% among those receiving inadequate care. After adjustment for measured covariates and targeting of the causal estimand, TMLE estimated a 14.2 percentage-point lower risk of stunting under adequate care. The direction and magnitude of this estimate were closely reproduced by g-computation and were consistent with the survey-weighted regression and inverse-probability-weighted analyses.

Several features strengthen the interpretation of this result. First, the effect was expressed on an absolute risk scale, which is directly interpretable for public-health planning. A difference of approximately 14 percentage points is potentially consequential at population level, particularly in a setting where stunting remains common. Second, the agreement between TMLE and g-computation suggests that the result is not strongly dependent on one parametric specification. Third, the association persisted when maternal healthcare was represented using alternative exposure definitions and when wasting and underweight were considered as alternative outcomes. Finally, complete-case and multiple-imputation analyses produced closely similar estimates.

The results are consistent with the broader nutritional context documented in Nepal. The 2022 NDHS showed that stunting varies substantially by residence, maternal education and household wealth (Ministry of Health and Population [Nepal] et al., 2023). Our analysis reproduces these socioeconomic gradients while extending the question from prediction or association to a marginal causal estimand. The BART findings are especially informative in this respect. The estimated benefit of adequate healthcare was larger among children of more educated mothers and in wealthier households. This pattern suggests that healthcare access may operate jointly with enabling social conditions. The effectiveness of a maternal-care strategy may depend not only on whether a woman reaches a health facility, but also on whether she can return for follow-up, understand health information, obtain adequate nutrition, and overcome geographic or financial barriers.

The urban-rural contrast provides a similar interpretation. Urban children had a more negative CATE than rural children, although both groups showed a benefit. Urban residence may provide greater physical access to facilities and a denser network of health services, but it is also a marker for other contextual characteristics. Because BART is estimating heterogeneity conditional on observed covariates, the subgroup pattern should not be interpreted as a simple geographic treatment interaction without considering residual confounding and selection into care.

The conventional odds ratios are useful comparators but require careful interpretation. The survey-weighted logistic model produced an OR of 0.58 and the IPW model an OR of 0.55. These estimates are broadly compatible with the TMLE result, but they answer a different question. Odds ratios are not equivalent to absolute risk reductions, and the odds ratio may differ from the marginal risk ratio even when the underlying causal effect is unchanged. The TMLE estimate therefore provides the

Table 7. Summary of principal analytical findings.

Target	Method	Key finding	Strength of evidence
Estimate causal effect	TMLE	ATE = -0.142 (95% CI: $-0.185, -0.099$)	Strong
Compare with conventional	LR, IPW	OR = 0.58 ($0.49-0.68$), OR = 0.55 ($0.46-0.65$)	Moderate
Heterogeneous effects	BART	Strongest in educated, wealthy, urban	Emerging
Sensitivity	E-value	2.18 for point estimate	Moderate
Survey design	Weighted	Results robust to survey weighting	Strong

more directly policy-relevant measure for this analysis.

The positivity diagnostics were reassuring overall, but the presence of a small proportion of observations with very high or low propensity scores indicates that some combinations of covariates were uncommon under one exposure level. Positivity is not merely a technical diagnostic: when treatment assignment becomes sparse in parts of the covariate space, the causal estimand becomes increasingly dependent on extrapolation (Petersen et al., 2012). The effective sample size of 2,456 compared with the nominal sample of 2,897 indicates that weighting reduced statistical information to some extent, although the primary estimate remained precise.

The E-value analysis provides an additional perspective on robustness. An E-value of 2.18 means that a single unmeasured confounder would need to be associated with both maternal healthcare and stunting by risk-ratio magnitudes of at least 2.18, conditional on the measured covariates, to move the point estimate to the null under the stated framework (VanderWeele and Ding, 2017). This represents a nontrivial confounding requirement, but it should not be interpreted as proof of causal validity. In particular, residual socioeconomic, nutritional, environmental and health-service factors may be correlated with both maternal healthcare and child growth.

An important limitation is that the present manuscript does not report empirical mediation estimates. The protocol proposed causal mediation through delivery and postnatal pathways, but the supplied results contain no direct, indirect or interventional mediation estimates. We therefore do not infer mediation effects from the total effect or from the alternative exposure analyses. Causal mediation requires additional assumptions concerning mediator-outcome confounding and exposure-induced mediator-outcome confounding, and these assumptions must be evaluated in relation to the temporal structure of the maternal-care pathway (Imai et al., 2010). Future analysis using the original NDHS recode files and a fully specified causal DAG should address this component explicitly.

The analysis also has limitations inherent to the observational survey design. Residual confounding cannot be ruled out, and the causal interpretation depends on exchangeability after adjustment. Maternal healthcare is multidimensional and may itself be influenced by unmeasured preferences, health-seeking behaviour, facility availability and prior pregnancy complications. Measurement error in maternal-care variables may attenuate or other-

wise distort the estimated effect. In addition, the survey is cross-sectional in design, so temporal ordering must be reconstructed from recorded pregnancy, delivery and child-history variables. Finally, the manuscript results supplied for this synthesis do not include the underlying Super Learner risk estimates, learner weights, mediation estimates or complete survey-design diagnostics. These omissions limit the extent to which the computational implementation can be independently audited from the manuscript alone.

Despite these limitations, the methodological triangulation is a major strength. The analysis combines design-based epidemiology with causal learning rather than treating machine learning as a substitute for epidemiological reasoning. The central result remains stable across regression, weighting, and TMLE, while BART adds evidence that the magnitude of benefit varies systematically across social strata. This combination is more informative than reliance on a single model because it separates the question of average effect from the question of who may benefit most.

CONCLUSION

In a nationally representative sample of Nepalese children, adequate maternal healthcare was associated with a substantially lower probability of stunting. The primary TMLE estimate indicated an absolute risk reduction of 14.2 percentage points (95% CI: -18.5 to -9.9), with closely concordant estimates from g-computation, inverse-probability weighting and survey-weighted regression. BART suggested that the estimated benefit was larger among children of mothers with higher education, children from wealthier households and children living in urban areas. These patterns point to the importance of combining maternal healthcare interventions with strategies that reduce socioeconomic and geographic barriers to effective care. The findings support the use of survey-aware causal methods for nationally representative observational health data, while recognizing that causal interpretation remains conditional on the underlying identification assumptions.

STRENGTHS AND LIMITATIONS

This study has several strengths, including the use of a nationally representative NDHS sample with survey-aware estimation, triangulation across regression, inverse-probability weighting, g-computation, and TMLE, and explicit assessment of treatment-effect heterogeneity us-

ing BART. Positivity diagnostics, sensitivity analysis for unmeasured confounding, and robustness checks across alternative exposure, outcome, and missing-data specifications further strengthen the credibility of the findings.

The principal limitation is the observational design, which precludes direct verification of key causal assumptions, particularly conditional exchangeability and the absence of important unmeasured confounding. Residual confounding and limitations in temporal ordering may therefore persist. Accordingly, the estimates should be interpreted as causal effects conditional on the stated identification assumptions rather than definitive causal effects.

DATA AVAILABILITY

The analysis is based on the 2022 Nepal Demographic and Health Survey. Access to DHS datasets is subject to the data-access conditions of The DHS Program and the permissions applicable to the investigators.

DECLARATIONS

Ethics approval: The study was taken consent to use secondary survey data for this manuscript.

Competing interests: No conflict of interest.

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